INCH-POUND

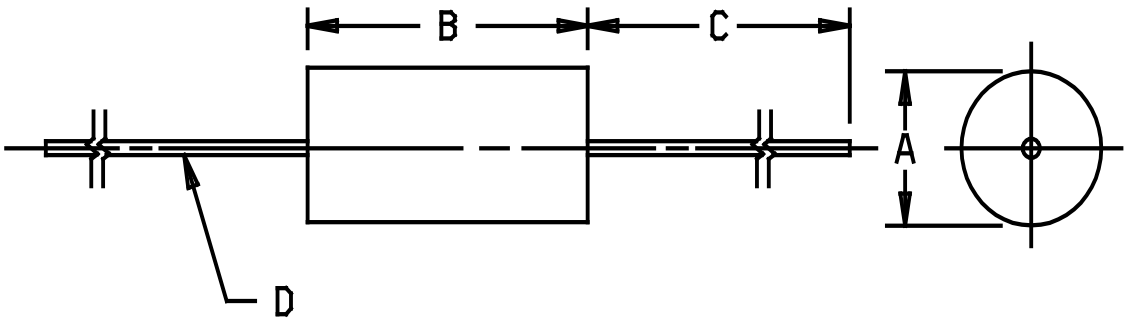
MIL-PRF-39010/1E
AMENDMENT 1
6 September 2018
SUPERSEDING
MIL-PRF-39010/1E
27 August 1997

PERFORMANCE SPECIFICATION SHEET

COILS, FIXED, RADIO FREQUENCY, MOLDED, MICROMINIATURE, MAGNETICALLY SHIELDED (PHENOLIC CORE-IRON SLEEVE), ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39010.



Letter	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.152 (3.86)	.172 (4.37)
B	.390 (9.91)	.430 (10.92)
C	1.250 (31.75)	1.626 (41.30)
D	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:29Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Material: Phenolic core with iron sleeve.

Weight: 0.03 ounce maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: MIL-STD-202-301; test voltage of 1,000 V rms.

Barometric pressure: MIL-STD-202-105, test condition C, 70,000 feet with a test voltage of 200 V rms.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

Percent coupling: 3 percent maximum.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part or Identifying Number (PIN): M39010/01 (dash number from table I).

Supersession data: This specification supersedes MS75087 dated 4 September 1985.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:29Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical characteristics (initial) and dash numbers.

Dash number 1/	Inductance μH	Inductance tolerance ± percent	Q minimum	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/
AR10**	.10	5, 10	50	25	250	.025	1,790
AR11**	.11	5	50	25	250	.034	1,530
AR12**	.12	5, 10	50	25	250	.034	1,530
AR13**	.13	5	50	25	250	.037	1,470
AR15**	.15	5, 10	50	25	250	.037	1,470
AR16**	.16	5	50	25	250	.047	1,300
AR18**	.18	5, 10	50	25	250	.047	1,300
AR20**	.20	5	49	25	250	.067	1,100
AR22**	.22	5, 10	49	25	250	.067	1,100
AR24**	.24	5	47	25	250	.11	853
AR27**	.27	5, 10	47	25	250	.11	853
AR30**	.30	5	46	25	250	.13	780
AR33**	.33	5, 10	46	25	250	.13	780
AR36**	.36	5	44	25	250	.18	670
AR39**	.39	5, 10	44	25	250	.18	670
AR43**	.43	5	44	25	235	.25	565
AR47**	.47	5, 10	44	25	235	.25	565
AR51**	.51	5	43	25	210	.33	490
AR56**	.56	5, 10	43	25	210	.33	490
AR62**	.62	5	42	25	190	.45	420
AR68**	.68	5, 10	42	25	190	.45	420
AR75**	.75	5	40	25	180	.59	370
AR82**	.82	5, 10	40	25	180	.59	370

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the product level (e.g., C, M, P, R, S) and will be added to the end of the dash number.

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:29Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5			-10
Group IV	±5	±(3% +.001 ohm)	2/	-10
Group V	±5	±(2% +.001 ohm)	2/	-10
Quality conformance inspection Group B				
Subgroup 1	±5	±(2% +.001 ohm)	2/	-10
Subgroup 3	±5			-10
Subgroup 4	±5	±(3% +.001 ohm)	2/	-10

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±(_ percent +.01 μH).
2/ The self-resonant frequency shall be not less than the value specified in table I.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39010, this document references the following:

MIL-STD-202-105, MIL-STD-202-301

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:29Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5950-2018-051)

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:29Z
Check the source to verify that this is the current version before use.